



MODULE 03 · 22 MIN

Project Context with CLAUDE.md

Stop re-explaining your stack. Write it once; Claude reads it every prompt.

Theory · CLAUDE.md is a behavior file (4 min)

`CLAUDE.md` lives at the repo root. Claude reads it **automatically on every prompt**.

It is a *behavior* file, not documentation. Every line must change Claude's output.

Five sections earn their place:

- **Stack** — languages, versions, frameworks.
- **Conventions** — naming, layout, lint rules.
- **Commands** — exact build / test / run / lint.
- **Do-not** — hard lessons, the traps.
- **Glossary** — domain terms only your team uses.

Trim test: delete a section; if Claude behaves the same, it was bloat. Aim **under 80 lines**.

CLAUDE.md at a glance

CLAUDE.MD · WHAT TO PUT IN IT

CLAUDE .md

Project shape

Stack, top-level dirs, how to run

Conventions

Naming, lint, error handling style

Tests

Where they live, how to run one

Don't

Files/areas Claude must not touch

Glossary

Project-specific terms in one line each

Five sections — **Stack** · **Conventions** · **Commands** · **Do-not** · **Glossary** — under 80 lines.

Reference · A lean CLAUDE.md (≤ 80 lines)

```
# CLAUDE.md
```

```
## Stack
```

- Python 3.11, standard library only.

```
## Conventions
```

- snake_case files; one command per module under cli/.
- Lint: ruff. Format: black.

```
## Commands
```

- Test: `pytest -q`
- Run: `python -m taskcli`
- Lint: `ruff check .`

```
## Do-not
```

- Do NOT add third-party deps without asking.
- Do NOT swallow exceptions; surface exit codes.

Reference · A complete CLAUDE.md (all 5 sections)

```
# CLAUDE.md – Notes API
```

Stack

- Python 3.11 · FastAPI · Pydantic v2 · SQLite (stdlib sqlite3).
- Tests: pytest + httpx. Lint: ruff. Format: black.

Conventions

- snake_case modules; routes in app/routers/, models in app/models.py.
- One Pydantic model per resource; never return ORM rows directly.
- HTTP status: 201 create · 200 read/update · 204 delete · 404 · 422.

Commands

- Test: `pytest -q`
- Run: `uvicorn app.main:app --reload`
- Lint: `ruff check . && black --check .`

Do-not

- Do NOT add deps without asking – stdlib + the four above only.
- Do NOT swallow exceptions; raise HTTPException with a clear detail.
- Do NOT write to the DB outside a repository function.

Glossary

- "note": {id, title, body, created_at} – body may be empty, title may not.
- "winner": the Best-of-N candidate chosen in Module 4.

Reference · Common mistakes

- Writing an `ABOUT.md` (documentation) instead of a behavior file.
- 200 lines of bloat instead of a lean 80.
- Skipping **Do-not** — and not committing the file (if it's not in git, it isn't real).

Live demo · Before vs. after CLAUDE.md (5 min)

The prompt — run it twice, unchanged (before, then after):

```
Add an `--export csv` flag to the task CLI that writes all tasks  
to a file. Match the project's existing conventions.
```

1. Run it on the Module 2 repo with **no** `CLAUDE.md` → off-convention output.
2. Drop in a 12-line `CLAUDE.md`; in a **fresh chat**, paste the **same** prompt → now follows conventions.
3. Trim test: delete one section, re-prompt, observe the drift.

Success signal: with `CLAUDE.md` present, Claude matches your naming/layout without being told.

▶ Your turn · Author your CLAUDE.md (10 min)

Exercise: `exercises/part-03/README.md`

Write a `CLAUDE.md` for your Module 2 repo (or a personal repo):

- All five sections: **Stack** · **Conventions** · **Commands** · **Do-not** · **Glossary**.
- **Under 80 lines.** Run the **trim test** at least once.

Prompt: *"Read this repo and draft a CLAUDE.md with Stack, Conventions, Commands, Do-not, Glossary. Keep it under 80 lines; every line must change your behavior."*

Success signal: on the next prompt, Claude obeys one convention you wrote — capture a proof screenshot.

✓ Done & next (1 min)

Definition of done

- ☐ [] `CLAUDE.md` < 80 lines, all five sections, committed to git.
- ☐ [] Trim test performed at least once.
- ☐ [] Proof screenshot of Claude obeying one convention.

Next — with rules in place, we generate *several* solutions and pick the best.

Module 4 — Build Faster with Best-of-N.